Customer Insights and Loan Process Optimization for a Multinational Bank

Company Overview:

A multinational bank operating across multiple countries provides services such as personal banking, corporate banking, and asset management. The bank has seen a decline in customers opting for personal banking services and aims to reverse this trend.

Business Use Case:

The bank wants to increase new customer acquisition, reduce attrition, and improve overall customer experience by leveraging its customer data more effectively. The lack of actionable insights from existing data, coupled with inefficiencies in manual processes, has resulted in:

- Delays in loan processing and onboarding, negatively impacting customer satisfaction.
- Limited ability to analyze trends in customer behavior for proactive retention strategies.
- Underutilization of customer data, hindering personalized offerings and competitive positioning.

The bank seeks a solution that enables efficient data extraction, centralized storage, and advanced analytics to address these challenges.

Current Setup:

- The bank has multiple documents containing customer information, such as bank statements, onboarding application forms, and loan application forms.
- Currently, underwriters manually retrieve and review these documents, leading to delays and inefficiencies.
- The lack of a centralized data pipeline prevents the development of machine learning (ML) or analytics solutions to improve customer experience.

Business requirements -

- Enhance the ability to gain actionable insights from customer data to increase acquisition and reduce attrition.
- Improve process efficiency for loan approvals and onboarding by reducing manual effort.
- Enable executives to access insights via dashboards to drive business decisions.

Technical requirements -

- Design a solution to extract structured information from various types of digital documents (multi-template, multi-region).
- Build a data pipeline to ingest, process, and store this information in a centralized database.
- Ensure ML models can be updated with new data as needed and retrained over time.

- Enable access to insights with low latency for bank employees like underwriters.

What's expected -

The candidate should consider this a real-world challenge and create a two-page write-up discussing their solution. They should focus on industry standards and logical assumptions to fill in any missing details. Clarifying questions can be addressed during the proposal presentation.

The write-up must include the following:

1. Problem Understanding
    a. Summarize the business problem and technical challenges being addressed.
2. Questions for the Client
    a. Identify clarifying questions related to business goals, data availability, scalability requirements, or other technical constraints.
3. Proposed Solution
    a. Provide a high-level summary of the proposed solution and its key benefits.
    b. Describe the overall approach, highlighting the machine learning and technical components.
4. Detailed Solution Architecture
    a. Include a diagram and explain the design. Candidates should specify cloud services used at each step or describe on-premise alternatives.
    b. Justify architectural choices (e.g., why certain services or frameworks were chosen).
5. Phase-Wise Plan
    a. Provide a clear roadmap broken into phases: PoC, pilot, and production.
    b. Include approximate timelines (e.g., weeks/months) and team size needed for each phase.
6. Metrics for Success
    a. Define key business and technical metrics that will demonstrate the success of the solution (e.g., time to process documents, customer onboarding time, insights delivered, model accuracy, customer retention rate).
7. Pros and Cons of the Solution
    a. List the key benefits, potential drawbacks, and risk mitigation strategies.